

PCCOE INTERNATIONAL GRAND CHALLENGE 2026 · PUNE, INDIA

THEME: AI FOR CLIMATE CHANGE · UN SDG 13: CLIMATE ACTION

HydroSentry-AI

Neural-Physical Earth System for Flash-Drought & Compound-Flood Resilience

Physics-Informed · Mass-Conserving · **Nugen Domain-Aligned**

AgriTech & Water Resilience

Drought & Flood Prediction

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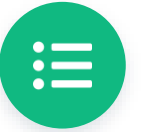
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Outline



Problem Statement & Idea

Stationarity is dead — a dual flash-extreme crisis

01



Objectives

Sub-hour, physics-true early warning & action

02



Proposed Solution & Architecture

Neural-physical engine + Nugen decision layer

03



Unique Features & Innovation

100× speed · zero-hallucination · 10-day foresight

04



Expected Outcomes

Deliverables, validation metrics & roadmap

05



Conclusion

Why HydroSentry-AI should win

06



Stationarity Is Dead — a Dual Flash-Extreme Crisis

Warming shattered the 100-year baseline. Two opposite disasters now strike the same basin — one in hours, one in days.



FLASH FLOOD · hours

Cloudbursts swell rivers and overtop levees in under 2 hours.



FLASH DROUGHT · 7–14 days

Heat & vapor-pressure deficit desiccate crop roots before any monthly index reacts.

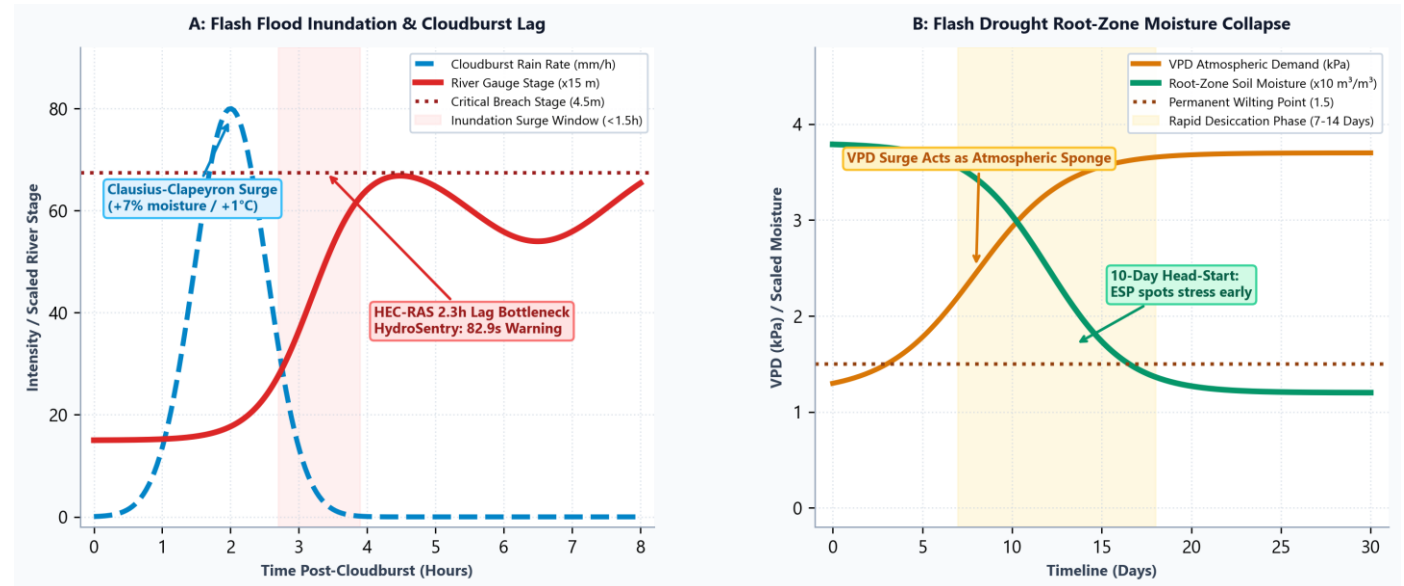
WHY EXISTING TOOLS FAIL

8,317 s

HEC-RAS TO SIMULATE (2.3 H — TOO SLOW)

+25%

BLACK-BOX AI WATER HALLUCINATED AT +4°C



Left: cloudburst → sub-hour river surge (HEC-RAS lags 2.3 h; HydroSentry warns in 82.9 s). Right: VPD surge collapses root-zone moisture — ESP flags stress 10 days early.



THE IDEA A physics-informed neural autopilot: mass-true flood & 10-day drought forecasts, turned into plain-language, life-saving directives by Nugen-aligned AI.



Objectives

KEY GOALS



Speed

Cut flood simulation 2.3 h → 82.9 s (100× real-time).



Physical trust

Guarantee mass conservation — zero water hallucination.



Foresight

Detect flash drought 10 days before visible wilting.



Autonomous action

Auto-issue certified, multilingual directives via Nugen.



Water balancing

Optimize dam FIRO pre-release: flood safety + summer supply.

END BENEFICIARIES



Farmers

pre-dawn drip alerts



Dam & water engineers

FIRO dispatch curves



Civic & disaster teams

geofenced evacuation



Citizens & agri-economy

protected lives & yield

ALIGNED TO UN SUSTAINABLE DEVELOPMENT GOALS

SDG
13

Climate Action

SDG
2

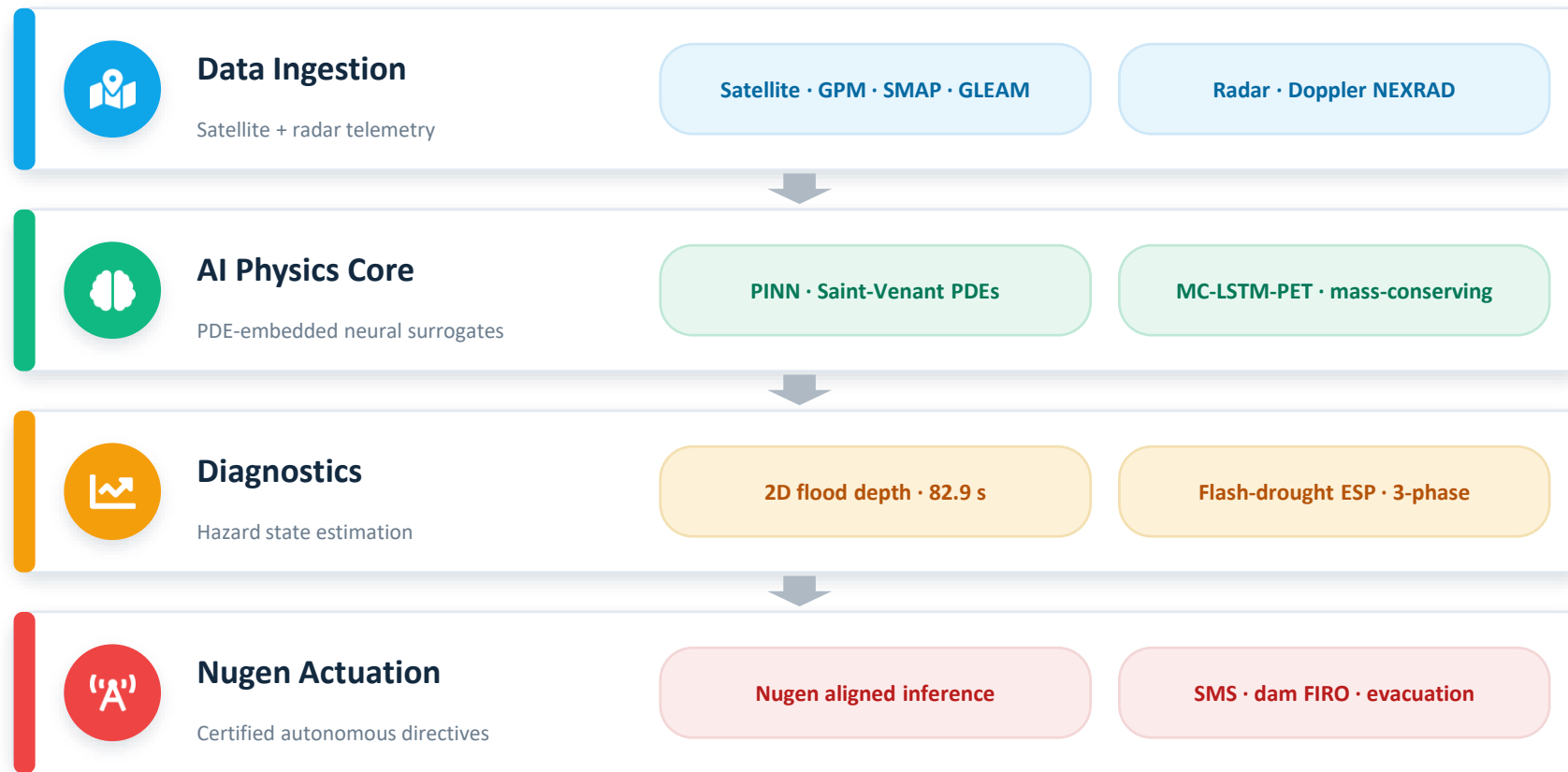
Zero Hunger

SDG
6

Clean Water



Proposed Solution & Architecture



SCALABILITY & FEASIBILITY

Basin-agnostic — retrains on any watershed

Serverless Nugen — zero GPU to operate

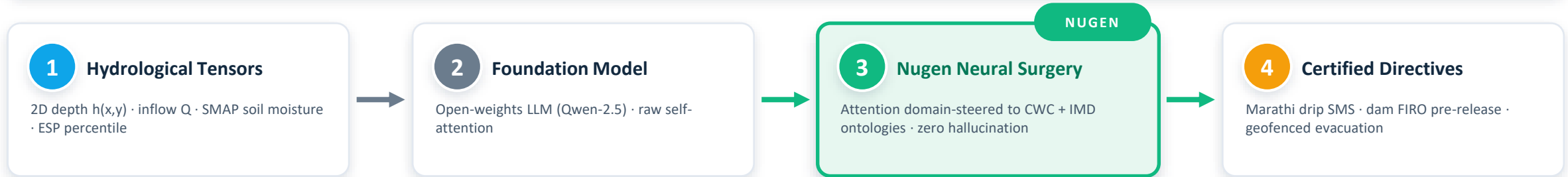
1 cloud GPU · 82.9 s real-time inference

TECH STACK

- Deep learning**
PyTorch 2.3 · torch.autograd (PDE residuals)
- Geospatial**
xarray · rioxarray · Py-ART · PostGIS
- Serving**
FastAPI · Uvicorn · Celery + Redis
- Decision layer**
Nugen aligned-model inference API
- Dashboard**
Next.js 14 · Mapbox GL / Deck.gl
- Alerting**
Twilio SMS / WhatsApp (multilingual)

Powered by Nugen Domain-Aligned AI

NUGEN CLOUD · POST /api/v3/inference · serverless · zero-GPU domain-aligned inference



What Nugen does

- “Neural surgery” domain alignment at train- & inference-time
- Constrains open-weights LLMs → zero hallucination
- Serverless, zero-GPU inference endpoints



How we call it

- POST /api/v3/alignment/create — align on CWC dam manuals + IMD rulebooks
- POST /api/v3/inference — hydro tensors → directives (temp 0.1)
- Base model: qwen-v2p5-0p5b-instruct



Autonomous directives

- Farmer: pre-dawn drip irrigation SMS (Marathi/Hindi)
- Dam: FIRO pre-release 450 m³/s, 25k ac-ft buffer
- Civic: geofenced levee-overtopping evacuation



Unique Features & Innovation



100× Real-Time Physics

PINN surrogate solves the exact shallow-water PDEs in 82.9 s vs 2.3 h — fast enough to evacuate.



Zero-Hallucination by Design

Hard PET ceiling + mass conservation: −7% drift at +4°C where black-box AI invents +25% water.



10-Day Drought Foresight

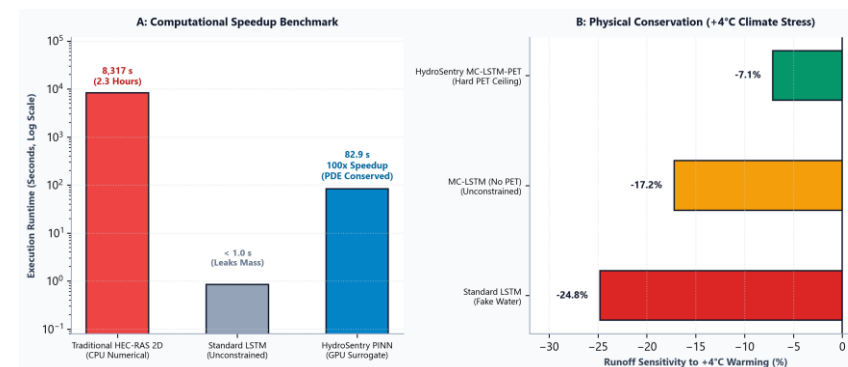
ESP reads NASA satellites to catch root desiccation before monthly SPI indices even react.



Autonomous Nugen Directives

First to fuse continuous PDE surrogates with a domain-aligned LLM for certified action.

BENCHMARKED PROOF



Left: 100× GPU speedup while conserving the PDE. Right: runoff drift under +4°C warming — physics keeps us honest.

0.93

KGE

0.57

DROUGHT POD

0.89

SM CORR. R



Expected Outcomes & Future Scope

DELIVERABLES



Neural-physical models

PINN + MC-LSTM-PET (PyTorch)



Decision API

FastAPI + Nugen inference



Geospatial dashboard

Next.js + Mapbox live overlays



Multilingual alerts

Twilio SMS / WhatsApp

VALIDATED IMPACT

100×

FLOOD SPEEDUP

0.93

KGE ACCURACY

0.57

DROUGHT POD

10-day

EARLY WARNING

R 0.89

SOIL-MOISTURE

FUTURE SCOPE · SCALING THE SENTINEL

NOW



Pilot — Upper Bhima / Pune

PINN flood + flash-drought ESP + Nugen directives, validated on one dipole basin.

NEXT



Scale — national, basin-agnostic

Re-trains per basin on live CWC/IMD feeds; Marathi/Hindi farmer app + dam FIRO desks.

HORIZON



Platform — multi-hazard, planetary

Adds cyclone storm-surge, GLOF & urban-flood modes; parametric crop-insurance triggers; on-edge AI.

Broken climate baselines, turned into **sub-hour, physics-true, life-saving action.**



Physics-true speed

100× faster than HEC-RAS — actionable in seconds.



Zero-hallucination trust

Mass conservation hard-wired into the network.



Nugen-aligned autonomy

Certified directives for farmers, dams & civic teams.

WHY SELECT HYDROSENTRY-AI



Scientific rigor: governing PDEs embedded, not merely learned



Mandatory Nugen inference — satisfied by design



Real regional grounding: Pune / Maharashtra basin dipole



Triple impact: SDG 13 Climate · SDG 2 Hunger · SDG 6 Water